ZCAS UNIVERSITY

BScCS41 - BACHELOR OF SCIENCE IN CYBER SECURITY/BScNE41- NETWORK ENGINEERING

CIT4351 CCNA CYBERSECURITY OPERATIONS

FINAL EXAMINATIONS

TUESDAY, 27th MAY, 2025

TIME: 14:00 – 17:00 HOURS

TIME ALLOWED: THREE (03) HOURS

INSTRUCTIONS TO CANDIDATES:

1. **SECTION A:** This section is **compulsory** with **ONE (01)** Question and carries **40 marks.**
2. **SECTION B:** Attempt **any THREE (03)** questions from this section. Each question carries **20 marks.**
3. This examination paper carries a total of **100 marks.**
4. Time allowed: **THREE (03) Hours plus FIVE (05) minutes to read through the paper.**
5. **DO NOT** turn this page **until** you are told to do so.

SECTION A – COMPULSORY QUESTIONS

INSTRUCTIONS: You MUST attempt **ALL QUESTIONS** that follow the case study provided.

QUESTION ONE

CASE STUDY: Ndalama Limited

Ndalama Limited is a rapidly growing digital payments provider based in Lusaka. The company recently suffered a multi-pronged cyberattack. First, several employees received phishing emails that mimicked the Central Bank's domain. Soon after, systems were infected with malware that ex-filtrated transaction logs to an unknown IP address. The company's security team, though equipped with some tools, failed to detect the attack until a customer reported missing funds. A review revealed that SIEM alerts had been ignored, and log files were not regularly reviewed. To strengthen its cyber defense, ***Ndalama Limited***'s management wants to adopt a Threat Intelligence strategy and invest in proper SOC tools and processes. Assume you are the SOC Expert at ***Ndalama Limited***;

a) Explain the importance of cybersecurity policies in ***Ndalama Limited*** and why it is critical for proactive network defense. **(06 marks)**

b) Describe **(04) four** common attack vectors that could have been used in this breach. **(08 marks)**

c) Explain **(03) three** real Zambian examples of incidents involving phishing, malware, DoS attacks, or insider threats. **(06 marks)**

d) Describe **(03) three** essential security tools used in a modern SOC environment that ***Ndalama Limited*** can adopt. **(06 marks)**

e) Compare Intrusion Detection Systems (IDS) with Intrusion Prevention Systems (IPS) in terms of function and deployment. **(04 marks)**

f) What is the CIA Triad in cybersecurity, and how could it guide ***Ndalama Limited***'s security posture moving forward? **(04 marks)**

g) Explain how SIEM and log analysis could have helped ***Ndalama Limited*** detect and respond to the attack sooner. **(06 marks)**

(Total marks for Section A: 40 marks)

SECTION B

CHOOSE ANY THREE (03) QUESTIONS FROM THIS SECTION

QUESTION TWO

CASE STORY: *TechZed Limited*

TechZed Limited, one of Zambia's main internet service provider, noticed a sharp spike in DNS traffic originating from a local corporate client. Upon investigation, their cybersecurity team found evidence of DNS tunneling, used by a threat actor to exfiltrate small pieces of encrypted data through DNS queries. This had gone undetected due to the lack of deep packet inspection and anomaly-based monitoring. As a Cyber Security Expert;

a) Explain what DNS tunneling is and why it is used by threat actors. **(04 marks)**

b) Describe **(03) three** signs in traffic logs that may indicate DNS tunneling. **(06 marks)**

c) Recommend **(02) two** tools that can detect and analyze DNS anomalies. **(04 marks)**

d) Suggest **(02) two** network-level controls that can prevent DNS abuse. **(04 marks)**

e) Describe **(02) two** SOC roles that would lead the response to this threat. **(02 marks)**

(Total: 20 marks)

QUESTION THREE

CASE STORY: *AutoBank*

AutoBank, has faced multiple reports of unauthorized withdrawals from customer accounts. Investigations by the internal IT team show that attackers may have exploited VPN credentials, likely harvested through a targeted phishing campaign. Logs indicate multiple failed login attempts followed by successful ones from unusual geographic regions. Additionally, endpoint protection reported suspicious activity on devices belonging to the customer support team. The bank has a SOC but lacks a formalized incident response plan. They have approached you to;

a) Explain the difference between a vulnerability and threat. **(02 marks)**

b) Recommend **(03) three** post-incident actions ***AutoBank*** should take. **(06 marks)**

c) What do you understand by the term "**evil twin**"? **(02 marks)**

d) Describe the **(03) three** main tiers of a SOC team. **(06 marks)**

e) Outline **(02) two** key log sources that a SOC analyst should review. **(02 marks)**

f) Identify **(02) two** possible indicators of compromise (IoCs) in this scenario **(02 marks)**

(Total: 20 Marks)

QUESTION FOUR

CASE STORY: *ZeleleMart*

ZeleleMart, a Lusaka-based online retailer, relies heavily on Microsoft 365 and Azure for daily operations. A misconfigured sharing policy in SharePoint allowed public access to confidential documents, including supplier contracts and staff payroll data. The breach went undetected until a supplier reported finding ***ZeleleMart*** files via a search engine. The company had not enabled audit logging or data loss prevention (DLP) policies. How can you;

a) Recommend **(03) three** measures for user awareness training to prevent cyber threats at ***ZeleleMart***. **(06 marks)**

b) Explain the concept of **Zero Trust** and how it applies to ***ZeleleMart***. **(04 marks)**

c) Identify **(02) two** Microsoft tools that help detect cloud security threats. **(04 marks)**

d) Differentiate between full packet capture ad metadata analyses. **(04 marks)**

e) Describe the role of NetFlow in network security. **(02 marks)**

(Total: 20 marks)

QUESTION FIVE

CASE STORY: *Malaiti Electricity Limited*

Malaiti Electricity Limited, one of Zambia's trusted electricity supply company, recently faced a wave of cyberattacks targeting its billing and SCADA systems. Threat actors used spear-phishing emails disguised as Ministry of Energy memos to deliver ransomware payloads that encrypted critical infrastructure files. The attackers demanded a ransom in Bitcoin. An internal investigation revealed that the cybersecurity unit lacked structured cyber threat intelligence (CTI) processes and had limited visibility of adversary behavior. There were also no threat feeds or sharing arrangements with other critical infrastructure organizations.

As a response, ***Malaiti Electricity Limited***'s board has mandated the development of a full-fledged Threat Intelligence Program and enhancement of its Security Operations Center (SOC) capabilities.

a) Describe Cyber Threat Intelligence (CTI) and explain its role in defending critical infrastructure like energy utilities. **(04 marks)**

b) Identify and explain the **(04)** four types of threat intelligence (strategic, tactical, operational, and technical). **(04 marks)**

c) Discuss how cyber kill chain or MITRE ATT&CK framework can help ***Malaiti***

Electricity Limited in understanding and stopping threats. **(4 marks)**

d) Describe any **(04) four** threat actors **(4 marks)**

e) Distinguish between cybersecurity and cybersecurity operations. **(4 marks)**

(Total: 20 marks)

END OF EXAMINATION